However, as under tunnelling, IPv6 datagram is repackaged for transmission over IPv4, it is done with an encryption standard to hide the nature of the original datagram while running through the tunnel.

Bootstrap protocol

Reverse address resolution protocol (RARP) is used to obtain the information of the IP address (logical address) corresponding to a given physical address of the machine. RARP does not provide the router or gateway address, server address, etc.

Bootstrap protocol (BOOTP) is used to provide additional information to a caller. BOOTP message goes as a payload of UDP layer. The message format is shown in Figure 2. Fields in the message are as follows:

- Operation field. Set 1 and 2, respectively, for request and reply
- Hop field. Set to 0 in the request message. When the message is passed from one server to another, hop count is incremented by one
- Transaction ID field. Used for sequential co-ordination between request and reply messages, just like the sequence number in IP datagram
- Seconds field. Used to count time in seconds, since the host starts BOOTP
- Others field. Self-explanatory; vendor-specific fields are not yet standardised

When a device (say, computer) connected to a network is powered up and boots its operating system, the system generates a broadcast BOOTP message. The message is a request for an IP address. A BOOTP configuration server assigns the IP address based on request. With the start of BOOTP procedure, when a request is sent to the server, a timer is turned on. If no reply is received within the defined time period, BOOTP should attempt retransmission. BOOTP is implemented with UDP as transport protocol. It operates only with IPv4 networks.

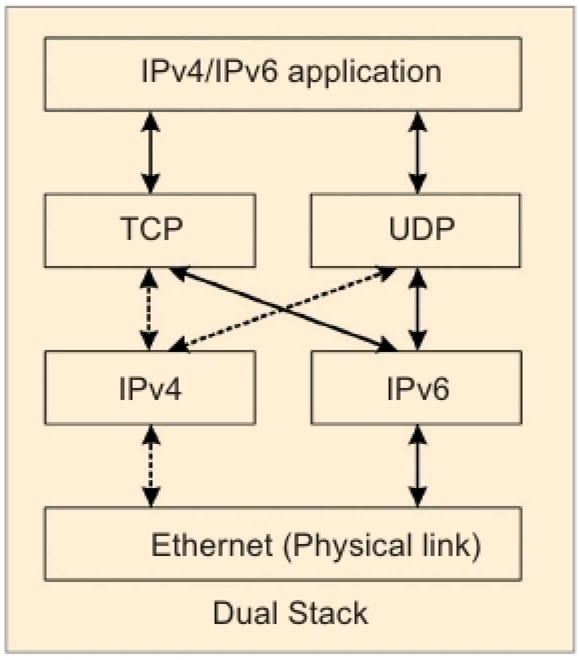

Figure 1: Illustration of dual stack

Dynamic host configuration protocol

IPv4 address is a stateful address. This means that, if a node moves from one subnet to another, the user has to either reconfigure the IP address or request for a new IP address from dynamic host configuration protocol (DHCP). DHCP is a network management protocol used over UDP/IP networks. A DHCP server dynamically assigns IP address and other network parameters to each of the devices connected to a network. With DHCP, an IP address is leased out to a device (host or computer) for a defined period of time.

Whereas, IPv6 supports a stateless auto configuration address. Thus, with IPv6, while moving from a subnet to another, a host can generate its own IP address. This is done by the host by adding its Media Access Control (MAC)/ physical address to the subnet prefix.

IPv6 also supports multiple addresses for each host. Three types of addresses may be made: valid, deprecated or invalid.

- With valid address, new and existing communication may be done.
- With deprecated address, existing communication may be done.
- With invalid address, no communication is done.

When a host attempts to get an IP address, a DHCP discover message (Figure 3) is broadcast over the physical

Operation (1 byte)
Hardware Type (1 byte)
Hardware Length (1 byte)
Hop (1 byte)
Transaction ID (4 bytes)
Seconds (2 bytes)
Client IPAddress (4 bytes)
Your IP Address (4bytes)
Server IP Address (4 bytes)
Router/Gateway IP Address (4 bytes)
Client Hardware Address (16 bytes)
Server Host Name (64 bytes)
Boot File Name (128 bytes)
Vendor Specific Field (64 bytes)

Figure 2: BOOTP message format

network with broadcast IP address 255.255.255.25. The discover message is received by all routers and other hosts, which they ignore. Only the DHCP server or router replies. Other routers discard the discover message. This prevents the entire Internet from being flooded with discover messages.

DHCP server in the network replies with DHCP offer message. The offer message provides an IP address and other information related to configuration. Every network need not have a DHCP server. A network may have DHCP relay agent, which requests the remote server for a reply to be subsequently sent to the requesting host. If the requesting host receives many offers from a number of DHCP servers, it will accept one offer and acknowledge the same to the corresponding server, which then acknowledges the reply of the host. Then, the host uses the IP address for communication.

Mobile IP

Mobile IP is the real next extension of mobile voice communication. Advances in wireless network technology have immensely changed various parameters like process, method, perception, quality and characteristics of voice communication. Consequently, developments in wireless data protocols have paved the way for mobile IP for data communication.